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Research Papers

Investigation on lithium-ion battery degradation induced by combined effect of current rate and operating temperature during fast charging

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ABSTRACT

Operating temperature and current rate are the main parameters that induce lithium-ion battery (LIB) degradation during the fast-charging process. In this study, fast-charging degradation was investigated using a commercial 18650 Nickel-Manganese-Cobalt battery at different charging current rates (C-rates) and operating temperatures. The degradation process was evaluated using electrochemical impedance spectroscopy and incremental capacity analysis. The electrode morphology change was investigated via a multiscale post-mortem analysis. The battery degradation mechanism at different C-rates and temperatures was characterized, and the coupled effects of the C-rate and operating temperature on the battery capacity fade were determined. With increasing operating temperatures, the most significant factor influencing the battery capacity fade changed gradually from lithium plating to the growth of the solid-electrolyte interface (SEI). An experimental analysis revealed that the battery capacity fade at the higher C-rates can be alleviated to some extent by employing higher operating temperatures. In comparison, the capacity fade at lower C-rates accelerated with increasing operating temperature. The changing and position shifting of the peaks in the incremental capacity curves were analyzed. It was found in the post-mortem analysis that the fast charging degradation was embodied in the macroscopic detachment of electrode material, the microscopic cracking of electrode particle, the SEI growth, the lithium plating, and the structural change of layered material crystal. The results of this study provide insight into the development of fast-charging strategies and the design of battery management systems.

1. Introduction

The development of electric vehicles (EVs) and hybrid electric vehicles (HEVs) has been widely promoted to limit the utilization of fossil fuels and reduce CO_2 emissions. Lithium-ion batteries (LIBs) have become one of the most important power sources for EVs and HEVs because of their high energy density and long cycle life and because they do not suffer from the memory effect. EV and HEV batteries must remain reliable for several years of use and must have a cycle life of more than 1000 cycles to ensure stable operation of the EVs and HEVs. However, in real-world applications, lithium-ion batteries experience severe degradation under harsh conditions, which inhibits the improvement of the EV mileage range and increases the cost of EV use.

Battery aging can be categorized into calendar aging or cycle aging. Calendar aging is the irreversible proportion of the capacity lost during storage, while cycle aging occurs when the battery is either charging or discharging. Cycle aging is more relevant to the real-world application of LIBs than calendar aging. The most accepted mechanisms of cycle aging include the growth of a solid electrolyte interphase (SEI) [1] and increased electrode particle cracking [2,3]. During the cycling process, a volume change of approximately 10% can occur in the graphite anode due to the insertion and extraction of lithium ions during battery cycling [4]. Because this volume change causes the SEI film to crack, the lithiated graphite inevitably contacts and reacts with the electrolyte solution species during each battery cycle, thus consuming the electrolyte and lithium inventory [5–7]. The formation of cracks and new SEI results in the continuous decrease of the available battery capacity, thickening of the SEI film, and increased battery resistance. Battery cycle aging is significantly influenced by various operating conditions, such as operating temperature and charging and discharging rates. Generally, battery aging is aggravated by high operating temperatures and high charging rates. Determination of the aging mechanism is complicated because many environmental factors and utilization modes interact to generate different aging effects.

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LIBs demonstrate satisfactory performance at room temperature but exhibit rapid capacity fade at elevated operating temperatures. Amine et al. [8] demonstrated cycle performance of battery with $LiFePO_4$ cathode. The battery capacity losses reach 70% at 55 °C and 41% at 37 °C after 100 cycles; moreover, almost no fading occurred at room temperature. Vetter et al. [9] demonstrated that elevated temperature not only enhanced the kinetics of the insertion and removal of lithium to and from the host lattice but also altered the morphology and composition of the SEI layers. Deng et al. [10,11] performed the state of health estimation on cell with NCA, NMC and LFP cathode. Results show that battery degradation accelerates with higher temperature and current rate. High-temperature cycling introduces lattice defects and exposes graphite edges that can react with the electrolyte to form inorganic compounds, thus increasing the SEI layer thickness. The SEI layer on the graphitic electrode changes in both morphology and chemical composition during storage at elevated temperatures due to electrolyte decomposition.

The charge and discharge current rates (C-rates) also significantly impact battery degradation. The C-rate is defined as the charge or discharge current divided by the battery's capacity to store an electrical charge. Ning et al. [12] studied battery capacity loss at different discharge rates (1–3 C) and found that the largest internal resistance increment of 27.7% could be achieved at a discharge rate of 3 C and that the rate capability losses were proportional to the increase in the discharge rate. A higher discharge C-rate intrinsically results in accelerated capacity fade due to the greater heat release from the battery and the mechanically induced damage to the active battery material. For high-rate charging, the lithium plating is the most significant factor that induces battery degradation. At higher charging rates, the anode potential could fall below 0 V and trigger lithium plating due to substantial anode polarization, resulting in drastic capacity loss and safety hazards. Gallagher et al. [13] conducted a battery cycling test at different charging rates and found that a cell with a 3.3 mAh/cm^2 areal loading had stable capacity retention upon cycling with a 1 C charging rate; however, the capacity rapidly faded at a rate of 1.5 C. In summary, battery degradation is promoted by an increased C-rate during cycling because the higher C-rate causes more mechanical damage to the electrode and increases the likelihood of lithium plating.

With the rapid development of EVs, fast-charging has become a critical issue for addressing the anxiety of EV owners about their mileage range. A higher C-rate (>1.5 C) is required to shorten the charging time, and the fast-charging process is a typical situation requiring a high C-rate. A high charging C-rate increases heat generation and internal battery temperature; therefore, the battery operating temperature significantly increases. A higher C-rate also directly accelerates SEI growth and lithium plating. Studies have confirmed that fast charging inevitably results in battery degradation due to the high C-rate. Therefore, combined effect of C-rate and operating temperature should be considered simultaneously. Preliminary investigations of the combined effects of operating temperature and C-rate have been conducted. Waldmann et al. [14] performed a comprehensive study on the effects of temperature on LIB aging by cycling 1.5 Ah graphite/NMC 18650 cells at a 1 C C-rate at different temperatures; they found that the cell cycle life was the longest at 25 °C and that higher or lower temperatures resulted in faster degradation. Ando et al. [15] tested commercial 18650 cells at 25 °C and 45 °C. For a 1 C charge rate, the cycle life was better at 25 °C than at 45 °C. At a charge rate of 2 C, however, the 45 °C cell had a significantly longer life than the cell at 25 °C. Yang et al. [16,17] developed a numerical model to simulate the temperature-dependent aging behavior of LIBs during fast charging and revealed that, for different charge rates, there is an optimal temperature that will provide the longest cycle life.

Based on this review, further challenges exist in the investigation of LIB fast charging. Most cycling tests have only used a charging rate lower than 3 C for testing the fast-charging performance. Furthermore, the combined effects of the temperature and the C-rate during fast charging have not been fully determined through experimental approaches. Therefore, performing a systemic study of the degradation processes of commercial cells during fast-charging is critical to revealing the degradation mechanisms of the batteries for different C-rates and temperature conditions.

In this study, commercial 18650 Nickel-Manganese-Cobalt (NMC) cells were used in a comprehensive evaluation of battery degradation during fast charging. The cycling test was performed under different charging C-rate and temperature conditions. The coupled effects of the C-rate and temperature on the battery capacity fade were analyzed using techniques that included battery cycling testing, electrochemical impedance spectroscopy (EIS), incremental capacity analysis (ICA). A multiscale post-mortem analysis was performed to evaluated the change of electrode material after fast charging cycling tests. The scanning electron microscope (SEM), energy dispersive spectroscopy (EDS) and X-ray diffraction (XRD) were employed to observe the change of electrode particle morphology, element maps and the electrode material crystal structure.

2. Experimental

2.1. Cycling test

A higher C-rate is necessary to achieve fast charging; therefore, a cycle-life test was performed at a high C-rate to determine its influence on the battery degradation process. The experimental system consisted of an environmental test chamber (NJ-40, precision: ±0.5 °C), a charge/discharge device (Arbin BT-2000), and an electrochemical workstation (Princeton, P4000). The environmental test chamber maintained the temperature during battery testing, and the battery charge/discharge device was used to perform the cycling test and ICA. The constant current-constant voltage (CC-CV) charging scheme and constant current (CC) discharging scheme were used in the cycling test. The voltage range of the cycling is 3.0–4.2 V. The cut-off current of charging process is 0.02 A. The time interval between each charging and discharging process is 30 min. An electrochemical station was used to perform the EIS test, during which a signal with a frequency between 0.01 and 100,000 Hz and an amplitude of 0.005 V was applied. The EIS tests were performed at an SOC of 100%. The lithium-ion battery used in this study was an NMC 18650 cell. The properties of the cell are provided in Table 1.

2.2. Test procedure

In the cycle-life test, two primary factors, i.e., charging rate and temperature, were considered. The testing procedure is shown in Fig. 1. The batteries were cycled at different operating temperatures and charging rates. Two groups of tests, G1 and G2, were performed during the experiment. A total of six batteries are used in the tests for two temperatures (25 °C and 40 °C). In G1 tests, three batteries were tested at 25 °C and each battery is cycled with a specific charging C-rate (1 C, 3 C or 5 C). In the G2 tests, three batteries were tested at 40 °C and each cell is also cycled with a specific C-rate (1 C, 3 C or 5 C). The discharging C-rates in G1 and G2 are both set to be 1 C. The rest time after each

Table 1
Battery parameters.

Manufacturer	LG
Size	18650
Style	Flat top
Cathode	$Li(Ni_{0.6}Co_{0.2}Mn_{0.2})O_2$
Anode	Graphite
Nominal capacity	2000 mAh
Continuous discharge rating	12 C
Nominal voltage	3.6 V
Dimensions	18.18 mm × 64.87 mm
Mass	48 g

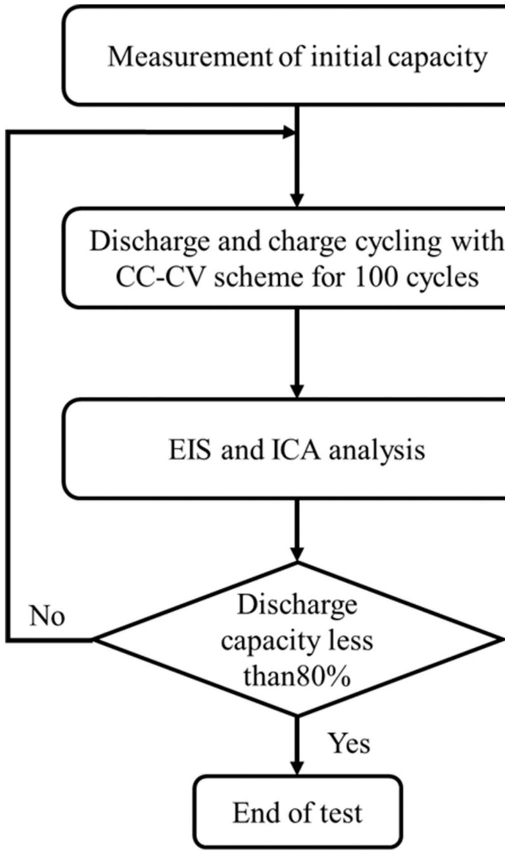

Fig. 1. Flow chart of the experimental procedure.

charging or discharging process is 30 min. When any battery reached 80% of its initial capacity, it was considered to be at end-of-life. The other batteries in the group were cycled to the same cycle number to ensure that all the batteries would experience the same degradation cycle time. In the G2 tests, three batteries were tested at 40 °C, using the same testing procedure as in the G1 tests. The EIS test and the ICA were performed on the aged batteries every 100 cycles. After the cycling test, the aged battery was disassembled to conduct post-mortem analysis including the SEM, EDS, and XRD observation. The cells were discharged to 3 V prior to disassembly and are disassembled in the Argon glove box. For comparison, the fresh cells were also cycled three times and disassembled to take the electrode samples. Prior to the post-mortem analysis, the electrode samples were soaked in dimethyl carbonate (DMC) solvent for 5 min to wash out the electrolyte. The particle size and morphologies were observed with the scanning electron microscope (Hitachi S-4800). The element mapping of the prepared samples was also obtained via scanning electron microscope (MAIA3 TESCAN). The XRD patterns were obtained on a Bruker DX1000 diffractometer. The XRD test was operated at 40 kV and 25 mA in the angular range of $2\theta = 10–90$ with an acquisition step of 0.02 and a scan rate of 0.2 per minute. The XRD data were analyzed using Jade software.

3. Results and discussion

3.1. Battery current and temperature variation during fast charging

The evolution of the battery-charging current was studied to determine the exact effect of the C-rate on battery-charging behavior. Fig. 2 (a) shows the battery current variations at different C-rates during one charging test at 25 °C. When the battery was charged at 1 C, the battery current curve first demonstrated CC, and then the battery current gradually decreased to the CV stage. The time percentages of CC and CV stages to the overall charging time are illustrated in Fig. 3. The current curve obtained at 1 C shows the typical current variation in the CC-CV charging scheme. When the C-rate increased to 3 C, the current curve still displayed the CC and CV stages, but the time percentage of CC stage decreased from 58.2% to 11.5%. The battery current reached 10 A immediately when the C-rate was increased further to 5 C, followed by a rapid current decrease to 5 A and a subsequent slow decrease. The CC stage disappeared in the 5 C charging test due to the high polarization of the battery. The battery voltage reached the charging voltage limit (4.2 V) at the beginning of the 5 C charging because of this greater polarization. The effective charging current was calculated to provide a quantitative comparison of the charging current at different C-rates. The effective charging current was defined as the arithmetic mean value of the gradually decreasing charging current. For the fast charging process, only the current values higher than 1 C-rate (2 A) were counted in the calculation. The calculation of effective charging current is shown in Eq. (1).

$$I_{\text{eff}} = \frac{\sum_{i=1}^{n} I_i}{n} (I_i \geq 2A) \tag{1}$$

where, I_{eff} is the effective current, I_{i} is the charge current at i th sampling point, n is the total number of sampling points. The effective charging

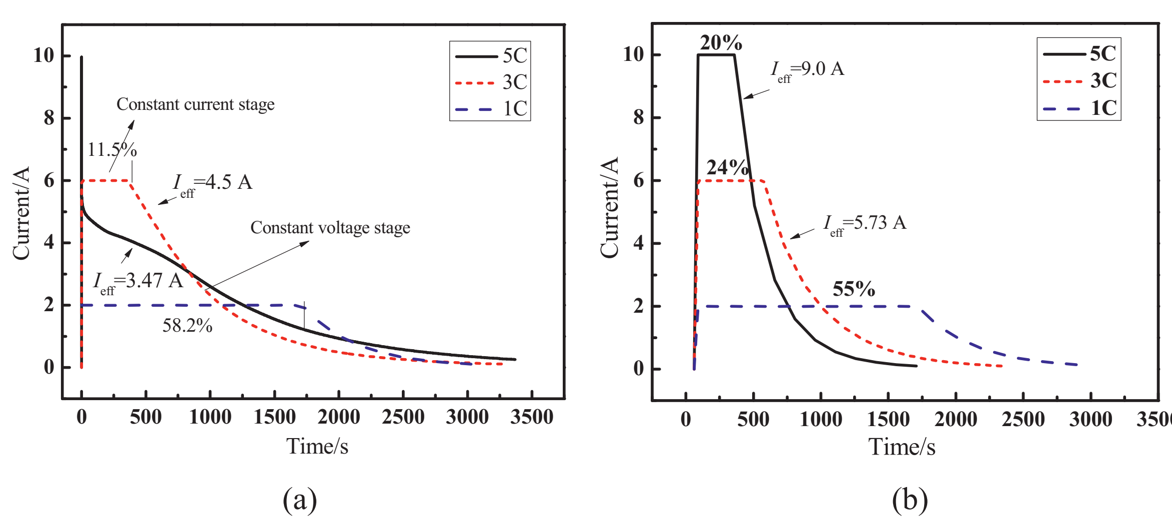

Fig. 2. Current variation at different C-rates and temperatures. (a) 25 °C; (b) 40 °C.

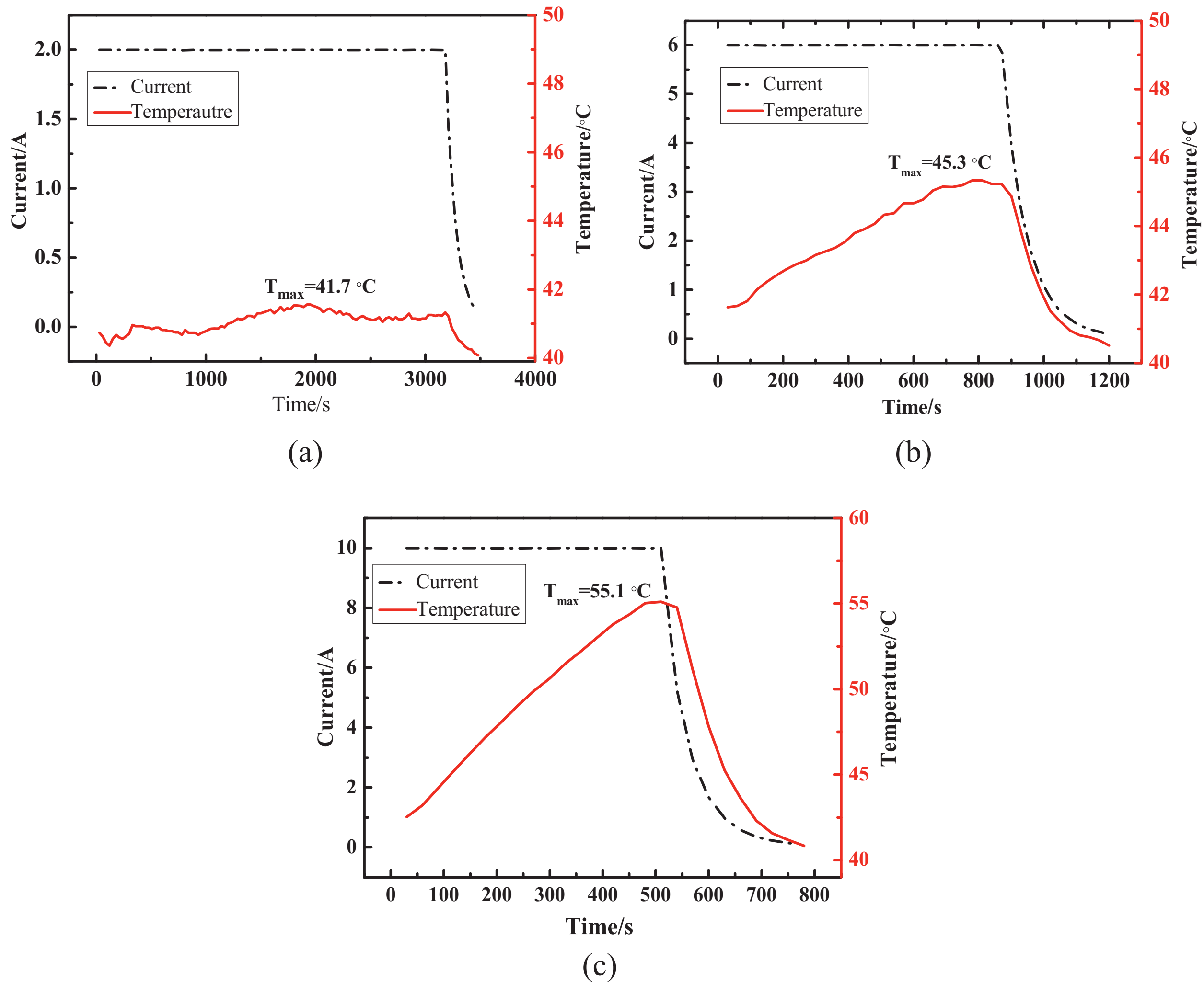

Fig. 3. The surface temperature variation of the battery during charging with different C-rates at 40 °C. (a) 1 C; (b) 3 C; (c) 5 C.

currents for the 1 C, 3 C, and 5 C C-rates were 2 A, 4.5 A, and 3.47 A, respectively. The effective current was significantly lower than the external applied charging current at the higher C-rates. The higher current rate induces an instantaneous higher surface lithium concentration at anode particle during the constant current stage. Then, the intercalation of lithium into the graphite is suppressed and the current is therefore vastly decreased at constant voltage stage. It is interesting that the effective current at 5 C is even lower than that at 3 C at 25 °C. This is attributed to the fact that the polarization is much higher for 5 C charging. Due to the limited lithium diffusivity in the solid electrode particle, the high polarization derived from constant 5 C charging still exists at constant current stage. This means that the battery current at constant voltage stage should maintain a much lower value to adapt the higher polarization reserved from the constant current stage.

Fig. 2(b) shows the battery current variations at different C-rates during one charging process at 40 °C. The processions of the CC stage were 20%, 24%, and 55% for the C-rates of 5 C, 3 C, and 1 C, respectively. The CC stages gradually decreased with increasing C-rates at 40 °C due to increased battery polarization. The effective charging currents for 1 C, 3 C, and 5 C charging were 2 A, 5.73 A, and 9.0 A, respectively. Table 2 compares the effective currents at different temperatures, showing that the effective current increased with increasing temperature. A higher effective current result in improved charging performance, which can be attributed to increases in mass transport and reaction activity. Specifically, the ion diffusivity and electrochemical reactivity increase with the battery temperature, which results in decreased internal resistance during the fast-charging process.

Table 2
Effective currents at different C-rates and temperatures.

25 °C	40 °C
4.5 A (3 C)	5.73 A (3 C)
3.47 A (5 C)	9.0 A (5 C)

When comparing the batteries cycled at different C-rates and temperatures, it is found that the CC stage observed at higher C-rate and temperature is larger than that observed at lower C-rate and temperature, such as 5 C cycling at 40 °C and 3 C cycling at 25 °C. This indicates that the influence of temperature is more significant than current rate in the case of this study. However, different LIBs show different temperature sensibility and rate capability due to different battery chemistry, geometry and manufacturing. Therefore, we could not make the conclusion that the variation laws of on the CC stage percentage and effective currents in this study are universally valid for all LIBs. The applicable scope of the results presented in Fig. 3 should be limited to the batteries with the same type and manufacturer as those used in this study (see Table 1), but results in Fig. 3 could provide evidences for analyzing the battery degradation patterns during fast charge cycling test.

In addition to the impact of the large charging current, the heat generation of fast charging process should also be evaluated, as the overheating of the battery is one of the main factors inducing the battery degradation. The evolutions of the battery surface temperatures during cycling at 40 °C were recorded to demonstrate the heat generation during fast charging. The surface temperatures of the battery during the charging process at 1 C, 3 C and 5 C were shown in Fig. 3. In the typical CC-CV charging process, the battery temperature increased in the constant current stage and decreased in the constant voltage stage. The maximum temperature increments of battery at 1 C, 3 C and 5 C were 1.7, 5.3 and 15.1 °C respectively. In 1 C charging, the temperature increment of the battery was not obvious due to the lower charging current. The prominent temperature increments were observed for 3 C and 5 C charging. The temperature increase is induced by the joule heat of large charging current. The large heat generation during fast charging aggravates the degradation of electrode material, because the side reactions such as the SEI formation and growth are enhanced at higher temperature. The maximum temperature increment of the battery obtained at different operation temperature and C-rates were demonstrated in Fig. 4. The maximum temperature increment increased linearly with the C-rate at 40 °C. In comparison, the temperature increment for 3 C and 5 C were close at 25 °C. This is attributed to the fact the effective charging current at 5 C is close to that at 3 C.

3.2. Coupled effect of the C-rate and operating temperature on battery capacity fade

The capacity retention of the battery was recorded during high-rate cycling to evaluate the capacity fade of the battery. Generally, battery capacity fade is strongly related to the C-rate and operating temperature. A higher C-rate increases the stress in the electrode and induces cracks on the electrode material, which directly results in incremental increases in the battery resistance and creates new locations for SEI layer formation. Moreover, a higher C-rate during the charging process can induce lithium plating on the anode material because, with the higher charging current, the lithium ion cannot be accommodated quickly enough between the intercalation layers on the anode surface. Higher temperatures influence the battery in two opposite ways: they increase electrochemical reactivity, as discussed in Section 3.1, while also increasing the speed of the side reactions in the electrodes, thus accelerating the growth of the SEI layer. The thicker SEI layer increases the internal resistance of the battery and causes the loss of lithium ions in the electrolyte. The battery degradation process becomes more complicated when the combined effect of the C-rate and temperature are considered.

The results of the cycle-life tests for the different charging rates at

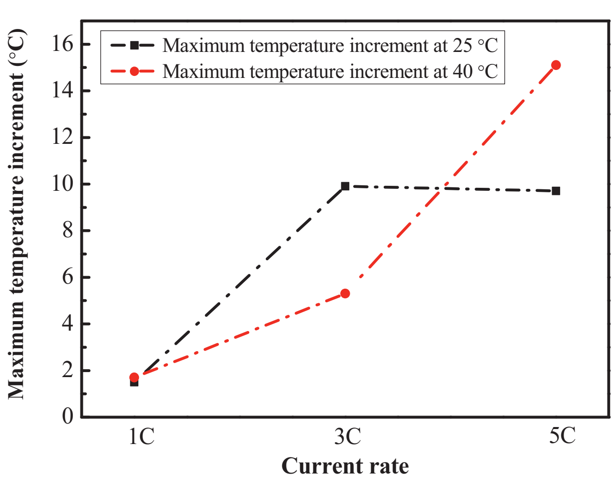

Fig. 4. The maximum battery surface temperature increment at different C-rates and temperature.

25 °C are shown in Fig. 5(a). The degradation rates for the different C-rates varied. The battery cycled at 1 C had the slowest degradation rate, losing only 7.2% of its capacity after 400 cycles. In comparison, after 400 cycles, the batteries with C-rates of 3 C and 5 C lost 25.1% and 18.5% of their capacities, respectively. At 25 °C, the battery capacity loss did not increase linearly with the increasing C-rate. This phenomenon is explainable for the CC-CV charging scheme because, at 25 °C, the effective current of the 3 C charging was greater than that of the 5 C charging. Therefore, it could be inferred that the battery degradation was more severe at higher effective charging currents. As discussed previously, a higher effective charging current induces the mechanical pulverization of the electrode material and lithium plating of the anode particles, resulting in increased resistance, the loss of active material, and the loss of lithium inventory.

The operating temperature was increased to 40 °C for the cycle-life testing to explore the influence of the operating temperature. The results of the cycle-life tests for the different charging rates at 40 °C are shown in Fig. 5(b). After 500 cycles, the batteries cycled at 1 C, 3 C, and 5 C lost 14%, 19%, and 30% of their capacities, respectively. The battery capacity loss increased linearly with increasing C-rate at 40 °C; this was in contrast to the results obtained at 25 °C. The trend of the capacity loss between 25 °C and 40 °C could be attributed to the effective charging current increasing linearly with the C-rate at 40 °C, which induced the linear variation of the capacity loss.

Comparing the degradation process of the battery at different C-rates at 25 °C and 40 °C reveals that the temperature significantly affected the battery capacity. Table 3 illustrates the battery capacity loss at the 400th cycle for different temperatures and C-rates. The battery degradation at 1 C increased with increasing temperature, while the battery degradation at the higher rates (3 C and 5 C) decreased with increasing temperature. The temperature increment had the opposite effect for low and high C-rates, primarily caused by the changes in the critical factors of influence at different temperatures and different C-rates. As indicated in the research of Waldmann [14], for a battery charged at 1 C-rate, the leading factor of battery degradation is the electrode lithium plating when the ambient temperature is less than 25 °C. The leading factor changes from lithium plating to SEI growth when the ambient temperature is higher than 25 °C. The theoretical research of Yang et al. [16] further analyzed the comprehensive effect of charging rate and temperature on battery degradation. It is pointed out that, the effect of lithium plating on the electrode during battery degradation increases with temperature under high-rate charging conditions. However, the capacity decrement caused by lithium plating is significantly higher than that caused by SEI growth. Based on the conclusions of the above theoretical research, it can be found that the effect of lithium plating of the electrode on the battery degradation at 40 °C is lower than its effect on the battery degradation at 25 °C. The reduced lithium plating effect at 40 °C alleviate the battery degradation under high charging rate. The experimental results verify the theoretical analysis conclusions of ref. [16], and it also shows that an appropriate increase in battery temperature could improve fast-charging performance.

3.3. Electrochemical impedance spectroscopy (EIS) study

Increased battery resistance is an important indication of battery degradation. In this study, EIS tests were conducted to quantify resistance variations. Typically, the EIS of the battery consisted of two semi-circles and one slope. The first semi-circle is observed at high frequency and was interpreted to represent the migration of the lithium ions through the multi-layered SEI. The second semi-circle occurred in the middle of the frequency range and was related to the charge transfer resistance across the interface. At the lower frequencies, the sloped line represents the solid-state diffusion of the lithium ions in the bulk electrode material. The Nyquist plots of the battery EIS are recorded every 100 cycles for different C-rates (1 C, 3 C, 5 C) and different operation temperatures (25 °C and 40 °C). Given the limited space, only one set of

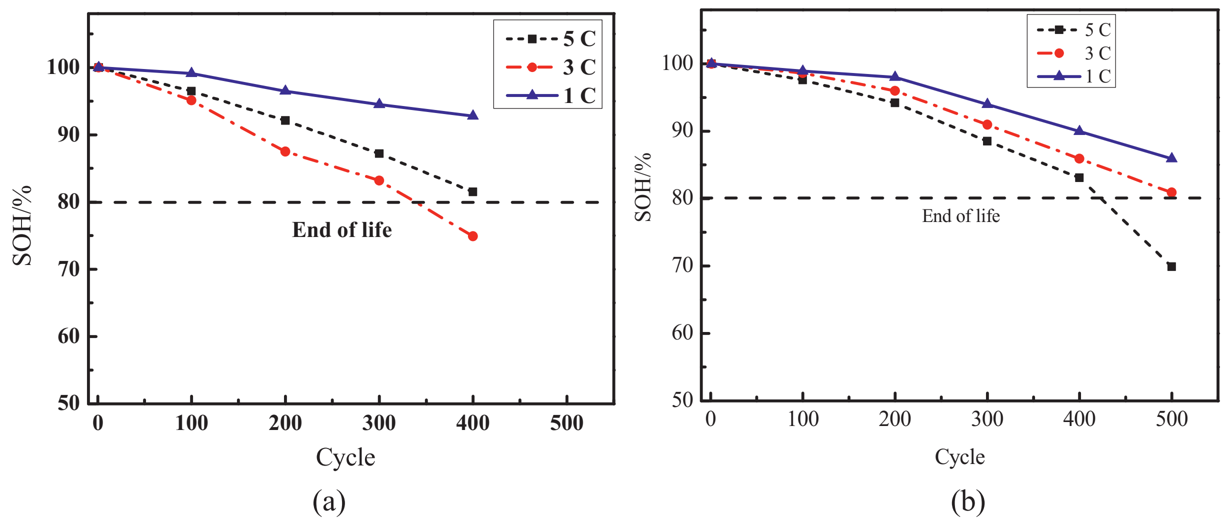

Fig. 5. Battery degradation for different C-rates and temperatures. (a) 25 °C; (b) 40 °C.

Table 3
Capacity loss after 400 cycles.

Temperature	Capacity loss (1 C)	Capacity loss (3 C)	Capacity loss (5 C)
25 °C	7.2%	25.1%	18.5%
40 °C	10.1%	14.1%	16.9%

Nyquist plots were illustrated in Fig. 6. The Nyquist plots were obtained from EIS tests of the cell cycled with 5 C-rate at 40 °C. The variation on the shape of Nyquist plots during 500 cycles can be observed. During the cycling process, the first semi-circle gradually increased with the increasing number of cycles, indicating that the thickness of the SEI increased during the cycling. The second semi-circle, representing the charge transfer resistance, also increased due to reduced reactivity in the charge transfer process. The rest Nyquist plots were analyzed using equivalent circuit model in next part.

The equivalent circuit model was introduced, and the battery resistance variation was analyzed to quantify the impedance in the Nyquist plots. The equivalent circuit model included resistor-capacitor (RC) networks to characterize the transient battery responses with different time constants associated with the diffusion and charge-transfer processes. The structure of the equivalent circuit used in this study is shown in Fig. 7. The equivalent circuit consisted of ohmic resistance, SEI resistance, SEI capacitance, charge transfer resistance, double-layer capacitance, and Warburg impedance. The commercial software ZsimpWin was used to fit the EIS data, and the values of the ohmic, SEI, and charge transfer resistances were obtained.

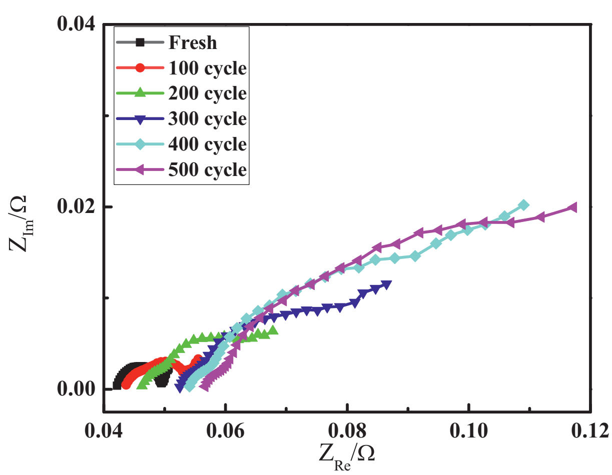

Fig. 6. Results of the EIS test for 5 C cycling at 40 °C.

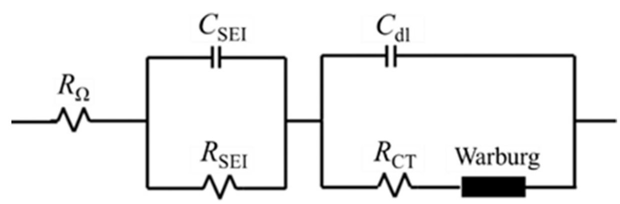

Fig. 7. Equivalent circuit of the battery.

The variations in the ohmic, SEI, and charge transfer resistance values under different charging rates at 25 °C were obtained and are illustrated in Fig. 8. Fig. 8(a) shows the changes in the ohmic resistance, which gradually increased with the number of cycles at different C-rates. The ohmic resistances is higher for higher charging rates due to the mechanical damage of the electrode material. The higher C-rate induced higher stress in the electrode, which resulted in more severe mechanical damage and a greater ohmic resistance increment [18]. As depicted in Fig. 8(b) and (c), the SEI and charge transfer resistances at different C-rates also increased as the number of cycles increased. The increases in the SEI resistance and charge transfer resistance were both caused by the growth of the SEI layer. The interface resistance continued to increase with the growth of the SEI, which caused the increase in the SEI resistance. Conversely, the growth of the SEI layer hindered the reaction process of the lithium ions on the surface of the active particles, which induced an increased charge transfer resistance. Comparing the ohmic, SEI, and charge transfer resistances in Fig. 8, the ohmic resistance values at the different C-rates were higher than the SEI and the charge transfer resistances, indicating that the resistance induced by the electron and ion conduction in the battery was significantly higher than the resistance induced by the SEI and electric double-layer charge transfer of the battery.

Fig. 9 shows the variations in the ohmic resistance, charge transfer, and SEI resistance during cycling at 40 °C. Fig. 9(a) shows the changes in the ohmic resistance with cycling, which were slight. In comparison, the charge transfer resistance and SEI resistance first changed slightly and then increased drastically after 400 cycles (Fig. 9(b)–(c)), which indicates that the increases of the SEI and charge transfer resistances were the predominant factors at 40 °C. The SEI layer on the anode grows faster and becomes more porous and unstable with high temperatures

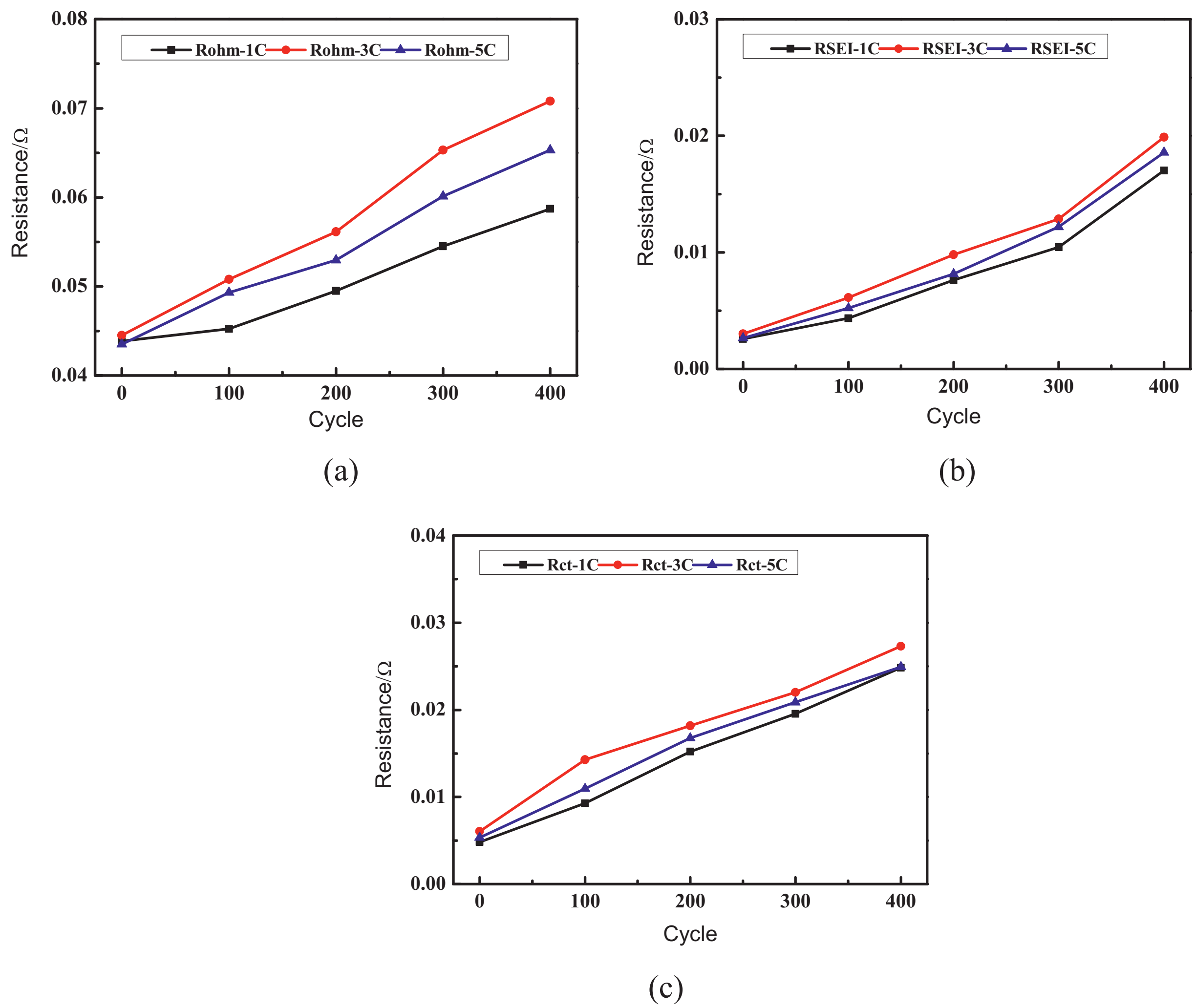

Fig. 8. Battery resistance variations during cycling at 25 °C. (a) Ohmic resistance; (b) SEI resistance; (c) Charge transfer resistance.

[19]. It is also found that the ohmic resistance values at the different C-rates were higher than the SEI and the charge transfer resistances.

3.4. Incremental capacity analysis

ICA is a method of examining the charge-discharge behavior of LIBs. ICA transforms the voltage plateaus on the charging/discharging voltage curve by differentiating the battery charging and discharging capacity versus the terminal voltage. Tracking the variations in the peaks could provide information about the health of the battery and the reactions occurring within it. The incremental capacity (IC) curves were obtained as follows: i) the 0.2 C discharge and charge cycle was applied to the test batteries; ii) battery capacity and voltage data were recorded at a sampling frequency of 1 Hz, and differential processing was performed on the collected capacity and voltage data to obtain dQ and dV; iii) the values of dQ/dV were plotted against the voltage V to obtain the IC curve.

As the battery was the most severely degraded at 500 cycles with 5 C and 40 °C cycling, the IC curves for this operating condition were selected for use in the ICA, and the corresponding IC curves during cycling were obtained. Fig. 10 shows the IC curves of the battery after different numbers of cycles. The IC curves represent both the charge and discharge processes. Fig. 10(a) shows that two peaks (A and B) could be identified in the IC curves of the discharge process. Peak A occurred at a voltage of 3.5–3.6 V, with an intensity of approximately 3 mAh/V. Peak B occurred at a voltage between 3.7 and 3.8 V, with an intensity of approximately 6.5 mAh/V. The peaks indicate the voltage at which the reactions occurred in the electrode. The reactions at the negative electrode primarily influenced peak A, and peak B reflects the sequential phase transition process of the nickel in the cathode material [20,21]. The intensities of peaks A and B both decreased with an increasing number of cycles, and the decrease in the peak intensity represents the loss of lithium in the active material [22]. The position of the peak shifted to a higher voltage range with an increasing number of cycles due to the increased internal resistance of the battery. Two peaks (C and D) were identified in the IC curves of the discharging process, as shown in Fig. 10(b). Peak C occurred at a voltage between 3.3 and 3.4 V with an intensity of approximately −2.5 mAh/V, while peak D occurred at a voltage of 3.5–3.6 V with an intensity of approximately −5 mAh/V. Peak C gradually vanished as the number of cycles increased due to the change in the transformation reaction at the graphite anode [23]. The intensity of peak D decreased with increasing numbers of cycles, and the position of peak D changed slightly. The increased battery internal resistance induced this peak position shift, while the loss of lithium inventory caused the decrease in the intensity of peak D.

The IC curves of charging process were compared for different C-rates and operation temperatures, as illustrated in Fig. 11. As shown in Fig. 11 (a), only peak D was clearly shown in the IC curves for different C-rates after 400 cycles at 25 °C. The peak intensity of IC curve of 3 C and 5 C are lower than that of 1 C cycling. The peak intensity of 3 C cycling is the lowest, which is attributed to the fact that the degradation at 3 C is the most severe. Similarly, as shown in Fig. 11 (b), the peak

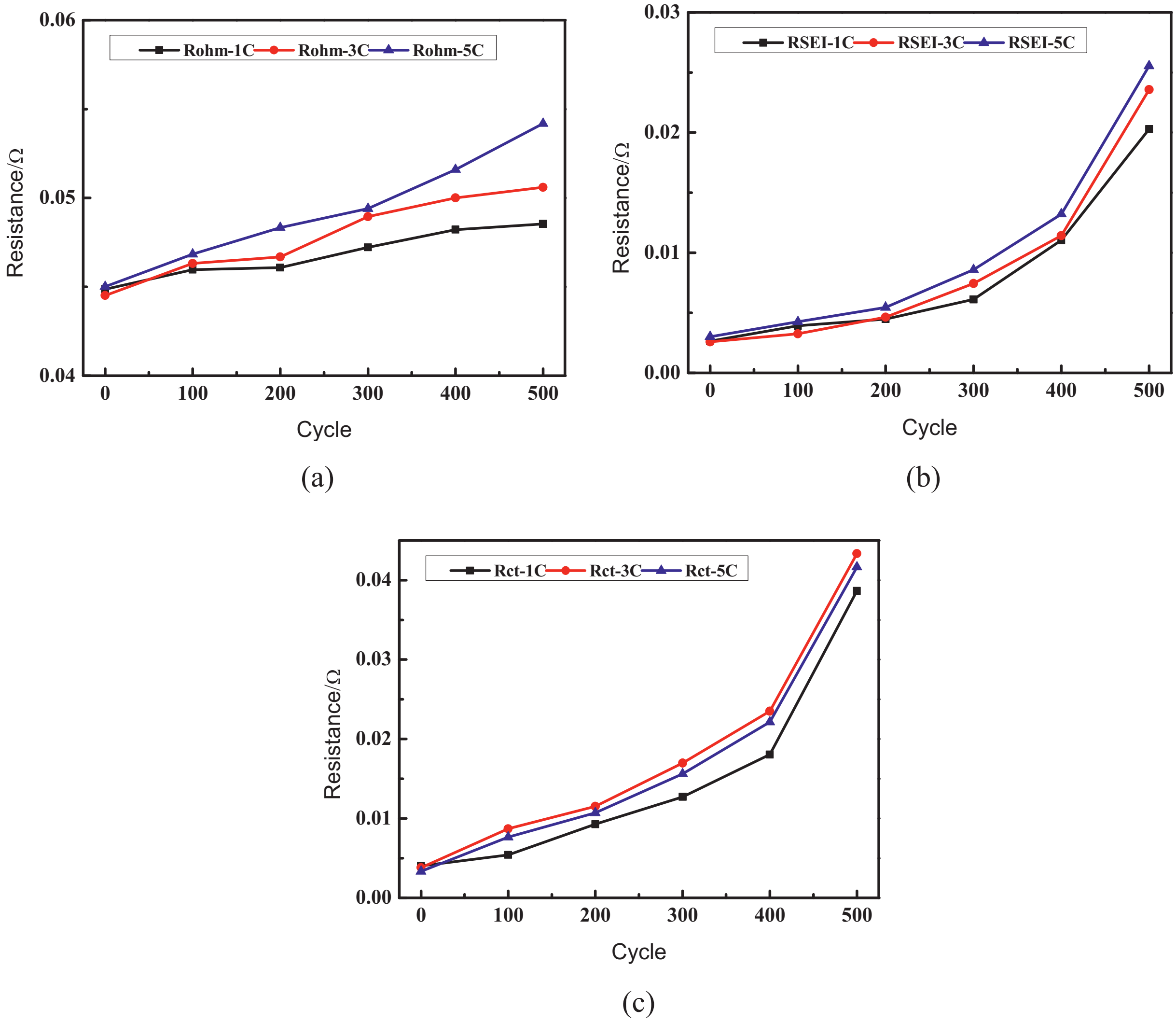

Fig. 9. Battery resistance variations during cycling at 40 °C. (a) Ohmic resistance; (b) SEI resistance; (c) Charge transfer resistance.

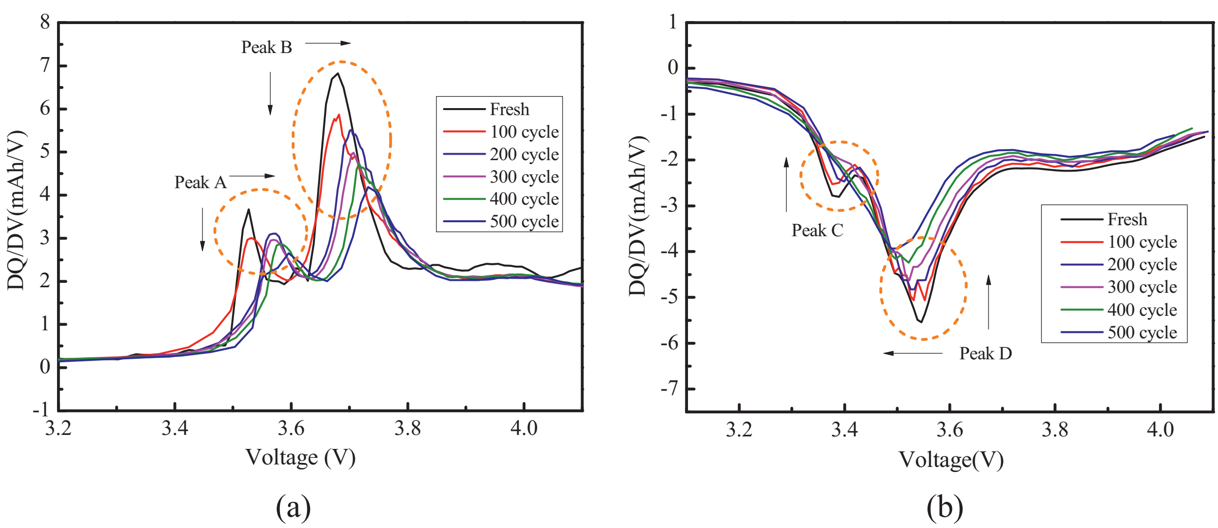

Fig. 10. Evolution of IC plots during 5 C and 40 °C cycling process. (a) Charge process; (b) Discharge process.

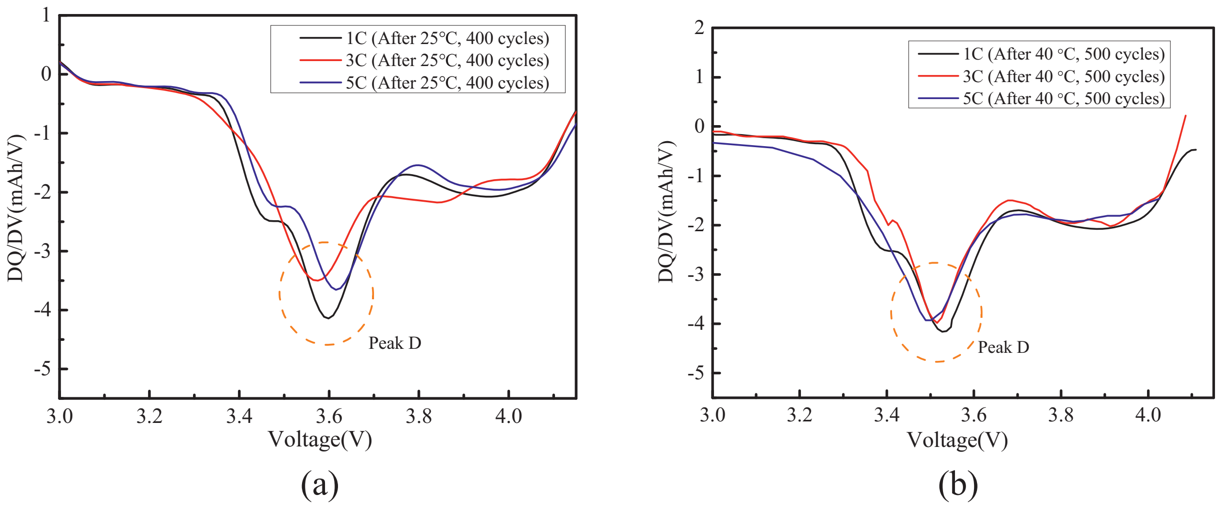

Fig. 11. The IC curves after cycling with different C-rates and temperature. (a) Charge for 25 °C cycling; (b) Charge for 40 °C cycling.

intensity of IC curve of 5 C cycling was the lowest at 40 °C due to its severe degradation. In addition, the positions of peak D are different for different C-rates and temperatures. However, we have not identified a fixed pattern of the variation of peak D position, as the positions did not change linearly with the increasing of current rates.

3.5. Post-mortem analysis via macro and micro observation

3.5.1. Anode electrode

After the cycling tests, the batteries were disassembled for the electrode morphology observation. The anode electrode sheets were harvested from the aged cell cycled with different C-rates and temperature for the macroscopic check. Meanwhile, the anode electrode material samples were collected from each anode sheets for SEM observation. The photos of anode electrode surface and the corresponding SEM were shown in Fig. 12. The anode electrode material samples for SEM observations were harvested from region without obvious damage (marked with red box) on the electrode sheets. Compared with the fresh cell electrode, the aged cell electrode suffered an evident detachment of anode material. The detachment was more severe for the cases with more severe degradation such as cycling at 3 C, 25 °C and at 5 C, 40 °C. This change of the electrode surface can partly

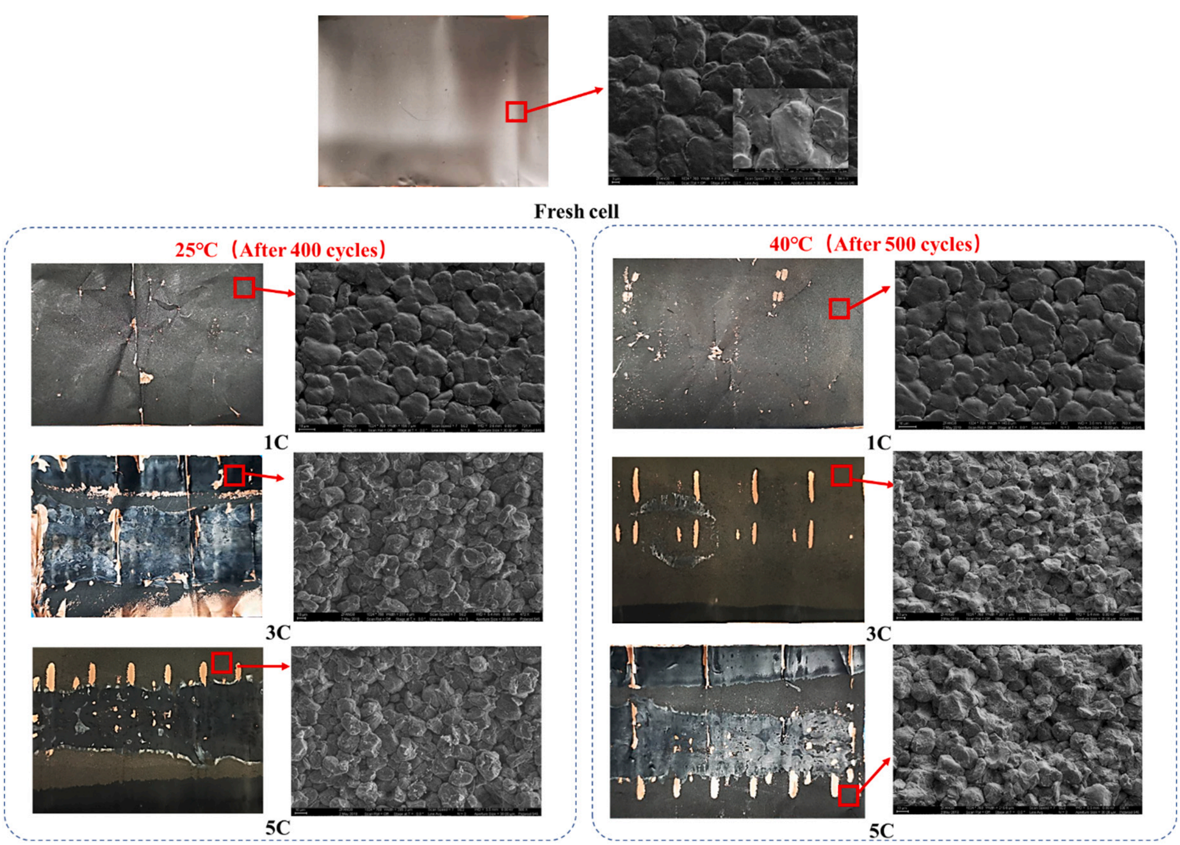

Fig. 12. Electrode surface and SEM observation of anode material.

account for the resistance increment during the cycling process. It is also found that the negative electrode, especially after 3 C, 25 °C and 5 C, 40 °C testing, exhibits areas of silver-gray color. These areas are considered to be cover by the products result from lithium plating in the anode material [24–26]. In addition to the macroscopic electrode surface, the microscopic morphology of the anode particles changed with battery degradation. For the fresh cell, the anode particles were tightly stacked, and the crevices between the particles were not prominent. After the high-rate cycling, cracks occurred on the anode surface. The cracks accounted for the observed increased battery impedance and decreased capacity. The cracks in the images of 3 C and 5 C cycling are much severe compared with 1 C cycling. However, the differences among the SEM images obtained at 3 C and 5 C were not obvious as all the electrode samples were harvested from the regions without obvious damage.

To further confirm the presence of lithium plating in the optical

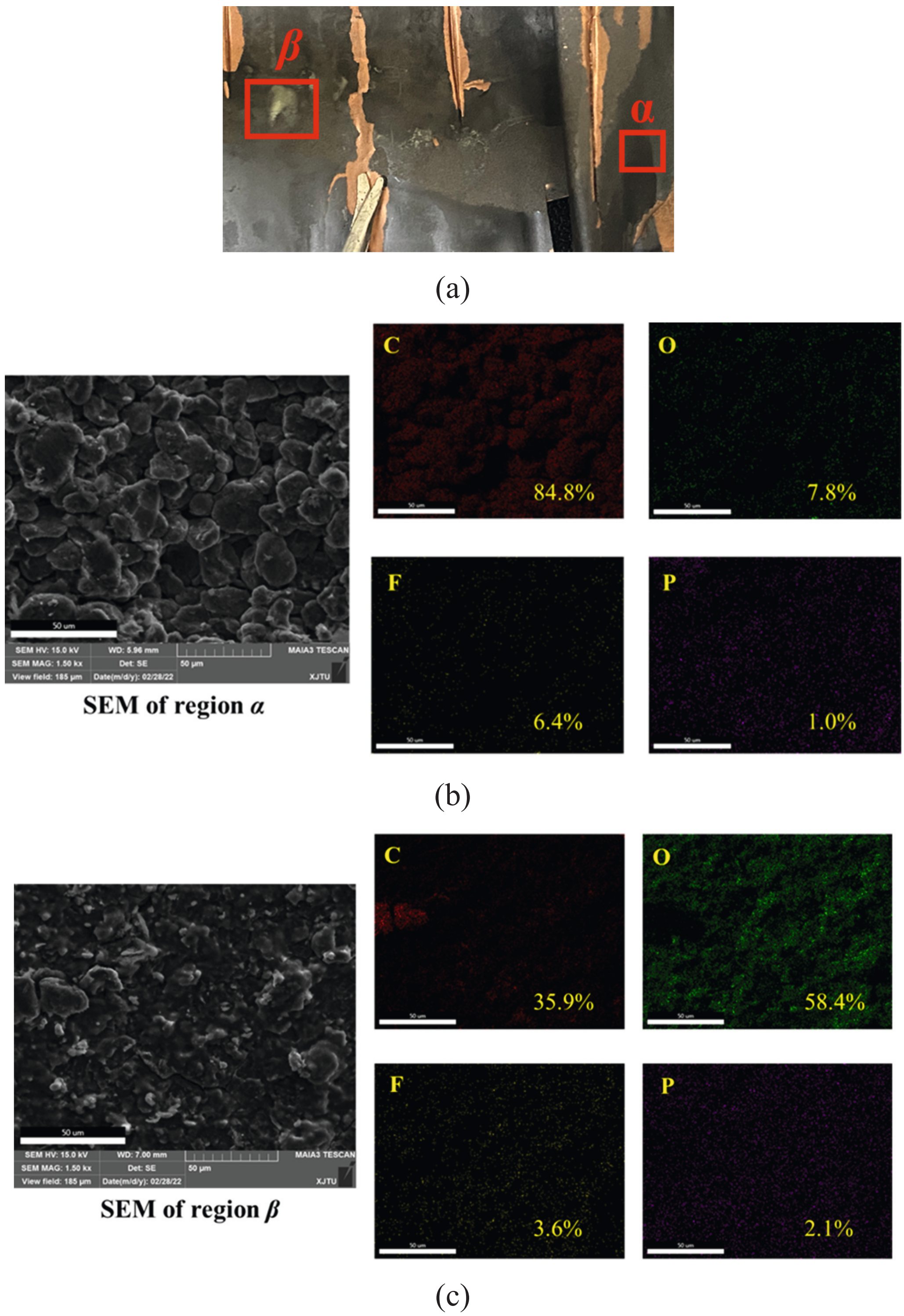

Fig. 13. SEM micrograph and elemental mapping of anode electrode after 3 C cycling at 25 °C. (a) Locations of region α and β; (b) SEM and elemental mapping of region α; (c) SEM and elemental mapping of region β.

electrode image observations, the SEM and EDS analysis was performed for an electrode sheet harvested from the battery after 3 C cycling at 25 °C, as the battery suffers the worst degradation at this condition. The electrode sheet showed an inhomogeneous plating behavior as some local regions presents lithium plating sites with silver-gray color. Therefore, electrode samples for SEM and EDS tests were collected from two typical regions, namely α and β, as shown in Fig. 13(a). The region α showed no obvious damage, and region β showed silver-gray color which indicated the occurrence of lithium plating sites. Fig. 13(b) shows the results of EDS analysis across region α where the presence of elemental carbon (red, atomic percentage, at.% = 84.8%), oxygen (green, at.% = 7.8%), fluorine (yellow, at.% = 6.4%) and phosphorus (purple, at.% = 1.0%) were obtained. Fig. 13(c) shows the EDS analysis across region β, and the content percentages are: elemental carbon (at.% = 35.9%), oxygen (at.% = 58.4%), fluorine (at.% = 3.6%) and phosphorus (at.% = 2.1%). Comparing the SEM figures and elemental mapping of two regions, it was observed that the oxygen content in region β was much higher than that in region α, and the film morphology was formed on region β. The significant high oxygen content suggests that the film observed in region β is induced by oxide formation on plated lithium during cell disassembly and exposure to air [27]. Besides, the oxygen, fluorine and phosphorus contents in region α originate from the formation of SEI layer [28]. It can be concluded from the EDS analysis that the anode suffers both SEI growth and lithium plating during its degradation.

The XRD patterns were obtained to check the crystal structure change induced by battery degradation. Fig. 14 and Fig. 15 showed the XRD patterns of anode material after cycled at 25 °C and 40 °C, respectively. As shown in Fig. 14(a), the diffraction peaks at around 26° are attributed to (002) plane of graphite carbon. The enlarged photo of (002) diffraction peak was shown in Fig. 14(b). The (002) peaks gradually shifted to lower angle. According to Bragg's law, the variation of (002) peak indicated that the interlayer spacing of graphite anode was gradually enlarged [29]. The similar change of the (002) peak location was found in Fig. 15. Comparing the location of (002) peaks for different C-rates and cycling temperature, it could be found that the change of (002) peak angle was more obvious with more severe degradation.

3.5.2. Cathode electrode

The cathode electrode sheets were also harvested from the aged cell cycled with different C-rates (1 C, 3 C and 5 C) and temperatures (25 °C and 40 °C) for the macroscopic check. Meanwhile, the electrode material particle samples were collected from each cathode sheets for SEM observation. The photos of cathode electrode surface and the corresponding SEM were shown in Fig. 16. Compared with the anode electrode sheets, the cathode electrode sheets did not show obvious morphology change. For the electrode sheets harvested from different C-rates and temperature, the electrode surfaces were well preserved and no distinct detachment of cathode material occurred on the aluminum sheet. The microscopic change of electrode can be observed from the SEM photos in Fig. 16. A distinct particle cluster occurred on the cathode electrode, and the morphology of the particles was apparent. However, as compared with the anode particles, the changes in the cathode morphology were not as significant and occurred because high-rate charging significantly impacts the anode material during the lithium intercalation in the carbon electrode.

Fig. 17 and Fig. 18 showed the XRD patterns of cathode material after cycled at 25 °C and 40 °C, respectively. As shown in Fig. 17, three peaks, (003), (101) and (104), were observed for cathode material. Two structural changes in the XRD patterns were identified. First, the peak intensity ratio of (003)/(104) peaks was changed with different C-rates. The intensity ratio of (003)/(104) peaks is a sensitive parameter to the cationic distribution in the lattice of ordered layered material [30,31]. The higher degree of the cationic mixing is achieved with lower (003)/(104) intensity ratio [32,33]. The distinct low ratios (<1.2) of (003)/(104) peaks were observed in XRD patterns of the 3 C cycling at 25 °C (Fig. 17(a)) and 5 C cycling at 40 °C (Fig. 18(a)). This phenomenon means cationic mixing is higher in the electrode material with more severe degradation. Second, the (003) peak shifted to lower angle after the cycling, which can be seen from the enlarged (003) peak in Fig. 17(b) and Fig. 18(b). The change of the (003) peak location indicates a larger interplanar crystal spacing along (003) in the structure of cathode material [34].

4. Conclusions

In this study, the degradation of LIBs during fast-charging was evaluated using commercial 18650 NMC batteries, which were cycled at different C-rates (1 C, 3 C, and 5 C) and different operating temperatures (25 °C and 40 °C). The battery capacities remaining after different numbers of cycles were measured, and the capacity fade was analyzed using EIS and ICA techniques. The microstructures of the electrode particle were observed to clarify the changes in the morphology of the cycled electrode material. The results demonstrated that higher temperatures could enhance the charging performance of LIBs because the effective current of the high C-rate charging increased at higher temperatures. Considering the effective current and the capacity loss at different C-rates revealed that the battery degradation rate increased

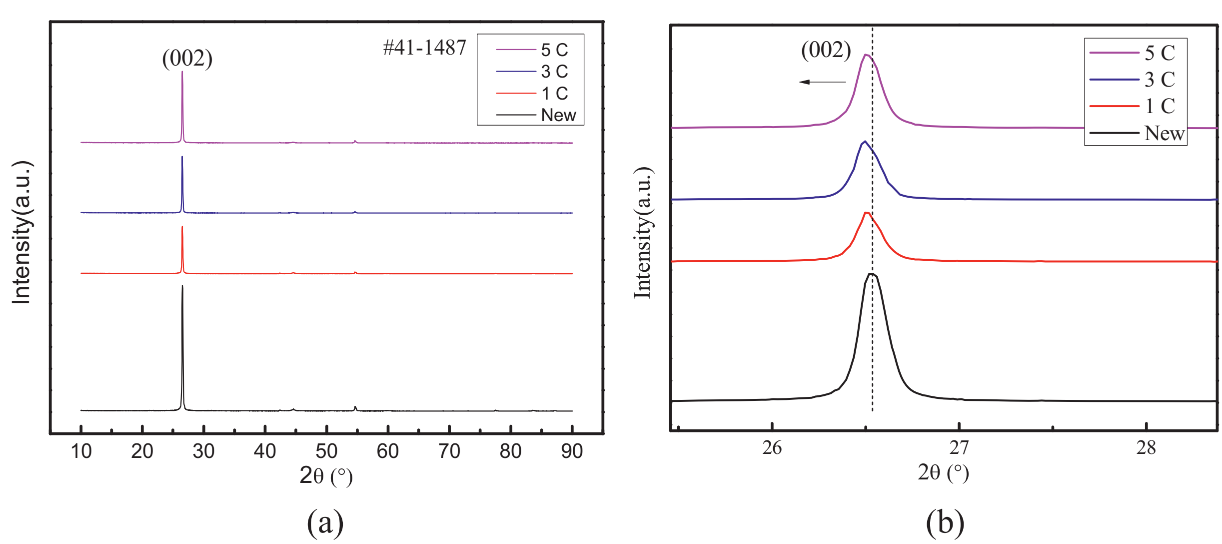

Fig. 14. XRD patterns of anode after 25 °C cycling with different C-rates. (a) Complete XRD pattern; (b) (002) diffraction peaks.

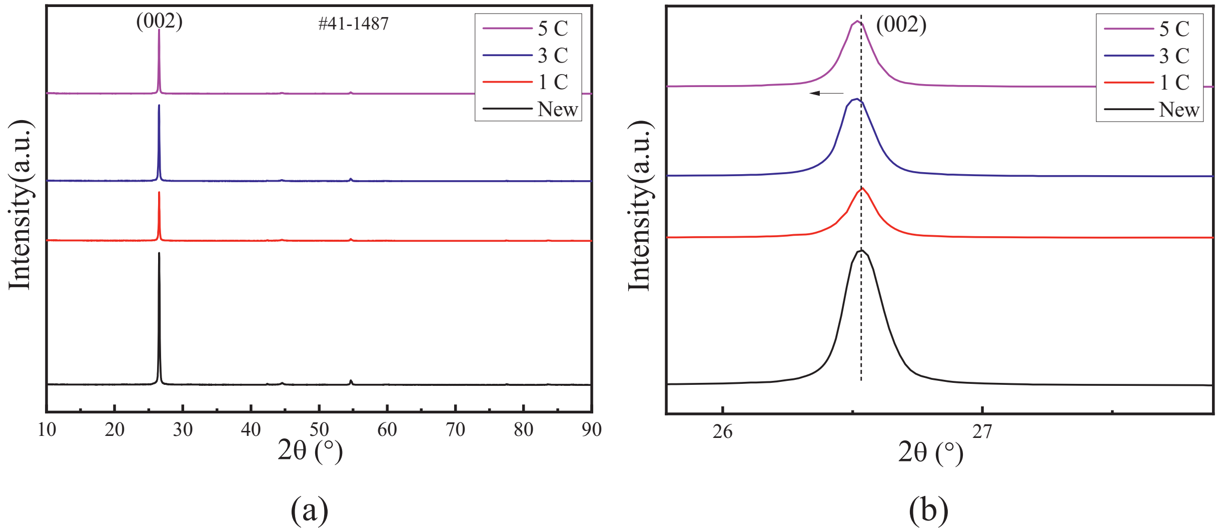

Fig. 15. XRD patterns of anode after 40 °C cycling with different C-rates. (a) Complete XRD pattern; (b) (002) diffraction peaks.

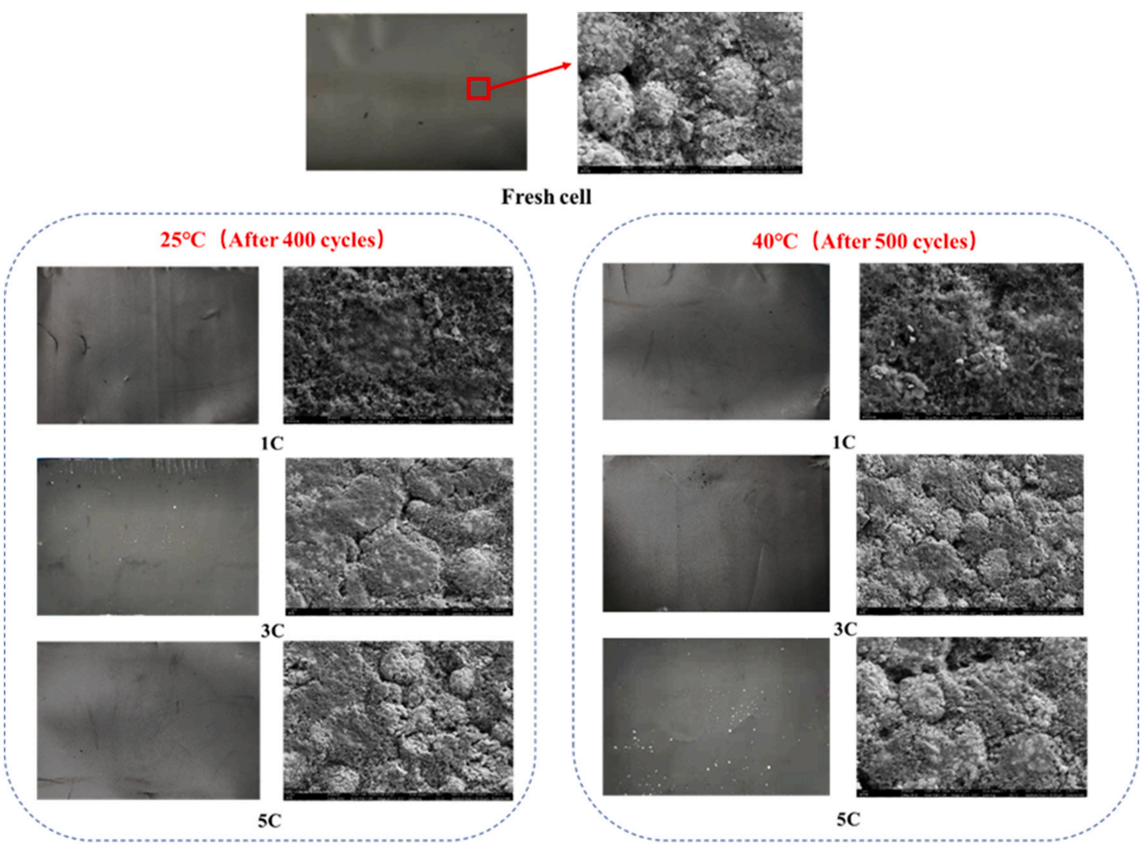

Fig. 16. Electrode surface and SEM observation of cathode material.

with higher effective currents. The EIS resistance values were similar, but the capacity loss was significantly different for different C-rates, which accounts for the occurrence of the lithium plating. Two peaks were identified in the IC curves of the charging and discharging processes. The intensities of the peaks gradually decreased during cycling due to the loss of lithium in the electrode material. The changes in the morphology of the anode particles were more significant than those of the cathode particles. The higher oxygen content observed in the elemental mapping of anode electrode sample with silver-gray region indicates the occurrence of lithium plating. The XRD patterns of cathode and anode materials were obtained. The (002) peak of the anode material and the (003) peak of the cathode material both shifted to lower angle due to the battery degradation at high C-rates. The low (003)/(104) peaks ratios were observed for the cathodes with severe degradation at 3 C (25 °C) and 5 C (40 °C) cycling. The results of this study could provide guidance to the fast charging protocol design for EVs by determining appropriate charging current rate and temperature environment.

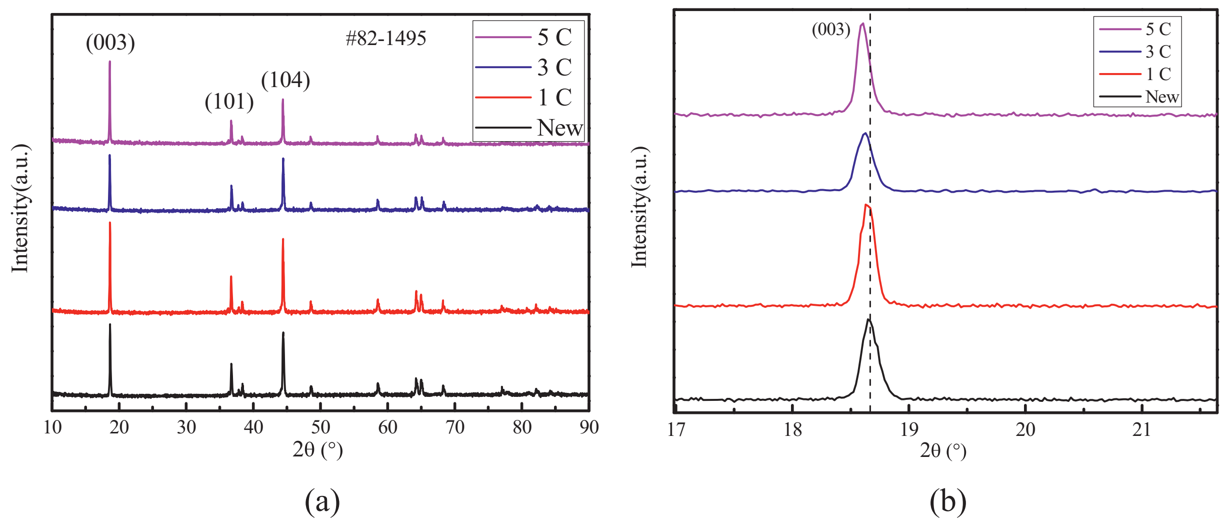

Fig. 17. XRD patterns of cathode after 25 °C cycling with different C-rates. (a) Complete XRD pattern; (b) (003) diffraction peaks.

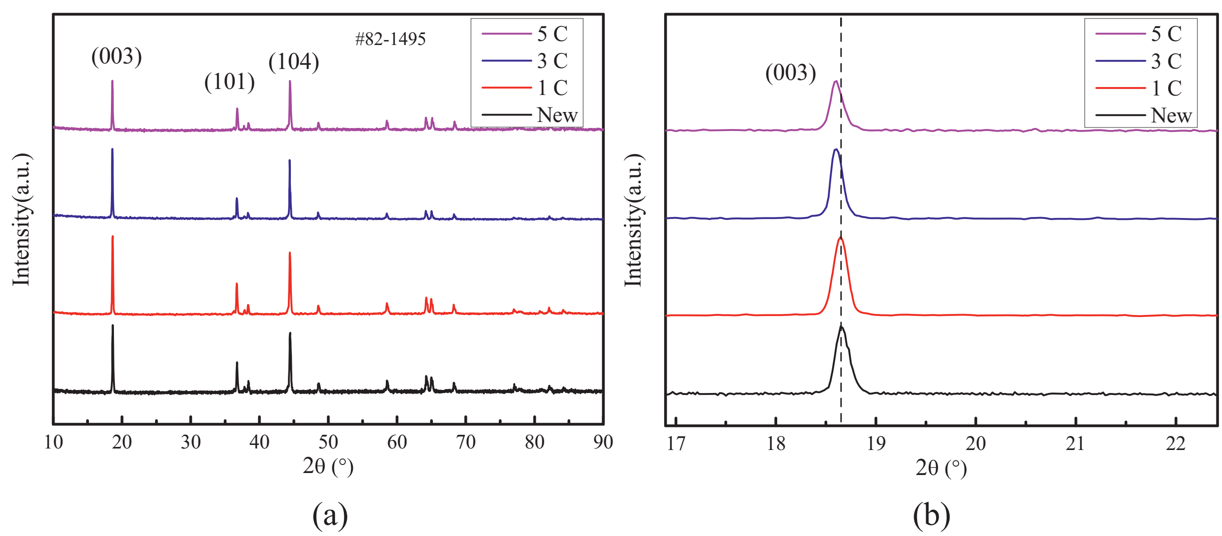

Fig. 18. XRD patterns of cathode after 40 °C cycling with different C-rates. (a) Complete XRD pattern; (b) (003) diffraction peaks.

CRediT authorship contribution statement

J.G. Qu: Conceptualization, Methodology, Investigation, Writing - Original Draft.

Z.Y. Jiang: Conceptualization, Supervision, Resources, Writing - Review & Editing.

J.F. Zhang: Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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